

Monotonic Search Networks for Computer Vision Databases

D. W. Dearholt, N. Gonzales, and G. Kurup
Computing Research Laboratory
New Mexico State University
Box 3CRL, Las Cruces, NM 88003
Telephone: (505) 646-3723

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Abstract

Design criteria for an associative network database for computer vision which supports monotonic search for the best match of an unknown entity are discussed. The network paradigm used provides for associative clustering of similar entities, and thus higher levels of abstraction are supported. A small database using thirty entities, each represented by a Fourier feature vector, is constructed and tested.

I. Introduction

This is a progress report on a project which was begun more than three years ago. It began with the belief that the organization of entities in a database according to objective properties of those entities, computed in a new way from distance measures, could support efficient search, classification, and clustering; and therefore higher levels of abstraction would be available, so that descriptions of the classification decision processes would also be supported. While associative storage and retrieval aren't new, there have been limitations in both theoretical aspects and in implementations. No system yet implemented, to the authors' knowledge, fully supports efficient search, clustering, classification, and higher levels of abstraction. Pathfinder networks (abbreviated PFNETs; Dearholt, Schvaneveldt, and Durso, 1985, and Schvaneveldt, Dearholt, and Durso, 1988) possess some of the properties of interest. Any PFNET contains all of the information of a hierarchical clustering scheme (Dearholt and Schvaneveldt, to appear in 1989). Furthermore, there are important connections between graph theory and PFNETs; two of the more relevant ones are that each Pathfinder network (1) contains all minimum-cost spanning trees, and (2) preserves geodesic distances between nodes (Dearholt, et al., 1985). However, as we developed and refined the theory of PFNETs, it became evident that they did not possess sufficient denseness of edges to support search without backtracking. Further investigation has led us to define a class of networks which supports search without backtracking, called *monotonic search networks* (abbreviated MSNET; in Dearholt, Gonzales, and Kirpekar, 1987). We did not generate an MSNET for the organization of the entities in the database in that paper, however, since it was a new concept. Instead, we used PFNET($L2, \infty, q=2$), which is a spanning subgraph of the MSNET, for the organization of the data. For some sets of data, this PFNET can provide monotonic search, but this is not guaranteed.

The databases of interest are composed of entities—those objects which are to be stored, recognized, and retrieved as needed by the vision system. Each entity is represented, we assume, by a feature vector. The objectives of this research are to develop a database for a vision system which supports search, classification of an unknown exemplar via template matching, and description of the decision processes leading to classification. Suitable domains for the database are those for which the entities to be stored are describable by features such as color or shape (perhaps using

Fourier descriptors for the latter). We assume that sufficient information is stored about each entity in the database to reconstruct a reasonably veridical representation of the entity. It is this feature which allows relatively complete description of the decision processes during search and classification. "Description" is used here to mean that the salient feature values of entities along the search path, the choices made which determine the search path, and higher levels of abstraction, are available for a summary of the entire process. Reasonable support of description requires that the identification of clusters (categories) is available in the database, and that the selection of the appropriate level(s) of abstraction is made evident by the query. We will not, however, consider the details of description further in this paper.

The Pathfinder networks, which are the predecessors of the MSNETs, originated as a model of semantic memory in humans. Many of the properties of PFNETs, previously called link-weighted networks (LWNs), are described in Dearholt, Schvaneveldt, and Durso (1985) and in Schvaneveldt, Dearholt, and Durso (1988). Because PFNETs provide for clustering of similar entities, they appear to be a good paradigm for some aspects of a database organization; indeed, their original purpose in modeling the semantic memory of humans provides for a database of concepts. Thus it is natural to extend PFNETs to feature-based applications in which each entity is described by a feature vector. Because the sparsest PFNET (and some of the denser PFNETs as well) is a spanning subgraph of an MSNET, the same properties of clustering and the provision for different levels of abstraction are supported by MSNETs. The efficiency in the search process is accomplished at the time the network is generated, by establishing appropriate edges. These edges provide a path between any two nodes so that the relative distance between nodes, as the path is traversed, is monotonically decreasing. Then, if a feature vector representing an external entity is presented to the database as a query, the corresponding node can be found rapidly from any node in the database. Heuristics to improve the selection of the initial node of the search can probably be developed to further improve the search efficiency.

Motivations for this database include (1) increased efficiency in the search process by eliminating backtracking; (2) the efficient determination (when appropriate) that a given entity is not represented in the database; (3) efficiently adding a node to the database when that is desired; (4) organization of the database so that similar entities are clustered together; (5) provision for category-level information by means of the clustering inherent in Pathfinder networks; and (6) support of description of the search and classification processes. It is these characteristics which we are regarding as our system specifications; when each of these properties can be supported, then the database will indeed provide the support desired for a robotics vision system. We have not yet devised an efficient algorithm for adding nodes to an MSNET, nor have we yet capitalized upon the structure of MSNETs in developing the descriptive aspects of the system. PFNETs and MSNETs are defined and discussed in Section II. The design and construction of the database system is described in Section III, and the set

of objects and their representation is described in Section IV. Results of the testing of the system are presented and discussed in Section V.

II. Theoretical Foundations: Pathfinder and Empty Neighborhood Networks

Development of the Pathfinder model has been ongoing for the past eight years. Pathfinder yields network structures (PFNETs) for a set of entities, given estimates or measures of the pairwise distances between the entities. In modeling human semantic memory, estimates of distances are usually obtained by first obtaining estimates of similarity provided by human subjects, and then subtracting each similarity value from a constant to compute the corresponding distance. For the database schema discussed here (for objective data), the entities are each represented by a feature vector, and the distance between a pair of entities is computed using the L_2 norm. The set of all distances between pairs of entities is presented to the PFNET generating system as a weight matrix. If the weight matrix is symmetric, then the PFNETs derived from it are undirected, whereas an asymmetric weight matrix yields directed PFNETs. We assume that there are no missing entries in the weight matrices used, and that the entries on the main diagonal are zero. For a weight matrix obtained using the L_2 norm on objective data, symmetry is guaranteed, and the network derived is undirected.

A complete network contains an edge between every pair of nodes, so that each weight in the weight matrix is represented by an edge. Such a network contains all the original information in the data, but it provides very little information about the structure underlying the data. Pathfinder networks, however, include only the edges necessary to preserve geodesic (minimum-distance) paths, thus facilitating analysis, clustering, abstraction, and interpretation. Two parameters are required for the complete definition of a PFNET for a particular weight matrix. These are the r -metric and the q -parameter. The r -metric is the value of the Minkowski parameter which is used to compute the distance between nodes in the network which are not directly linked. The distance between two nodes is computed as follows:

$$d(N_i, N_j) = \left[\sum_{k=1}^p w_k r \right]^{1/r}$$

where the w_k are the weights along the path between N_i and N_j . The r -metric parameter may take on values from 1 to ∞ . If $r = 1$, weights along a path are added to compute the total distance. If $r = 2$, the distance is computed using the familiar Euclidean metric. If $r = \infty$, the distance between two nodes along a particular path is the same as the maximum weight on the path. The q -parameter determines the maximum number of edges in paths considered to connect two nodes. For example, for $q = 2$, paths having more than two edges are not considered in the preservation of minimum-distance (geodesic) paths in the PFNET. The notation used for a PFNET in which pairwise distances between feature vectors are used to compute W is PFNET(Lm, r, q), with the parameters in

parentheses corresponding to the parameters discussed above.

The Pathfinder model leads to a type of clustering based upon the similarity of nodes. Indeed, hierarchical clustering using the single-link rule (minimum distance between clusters) can be obtained from a PFNET by means of the node sublists created during generation, although the converse is not true. The PFNETs, however, are themselves rarely hierarchical, in the sense that relationships between clusters are given explicitly by the edge structure, and are not limited to hierarchies. Thus PFNETs provide a means of scaling data similar in some respects to other clustering methods (e.g., Shepard and Arabie, 1979) and to multidimensional scaling (Kruskal, 1977), but the edges in PFNETs provide information not directly available in clustering or in multidimensional scaling. Another network scaling scheme is NETSCAL (Hutchinson, 1981),

but unfortunately Hutchinson did not consider triangle inequalities of dimension greater than two.

PFNETs possess properties of inclusion which vary as values of the r -metric and q -parameter change (Dearholt, et al., 1985). Briefly, for a particular weight matrix, PFNET2 is a spanning subgraph of PFNET1 if and only if the r -metric used for PFNET1 is less than or equal to the r -metric used for PFNET2, provided that the q -parameter is held constant. In addition, for a particular weight matrix, PFNET2 is a spanning subgraph of PFNET1 if and only if the value of the q -parameter used for PFNET1 is less than or equal to the value of the q -parameter used for PFNET2, provided that the r -metric is held constant. The PFNET generated with $r = \infty$ and $q = n-1$ (the-number-of-nodes-less-one) always has the minimum number of edges and is the union of all minimum cost spanning trees of an undirected PFNET.

Investigation of transformations on the values of the weight matrix has yielded two results of importance relating to the structure of PFNETs. A multiplicative transformation applied to the elements of a weight matrix preserves edge structure in the PFNET for any values of r or q . A monotonic transformation applied to data in a weight matrix preserves the structure of the PFNET only for $r = \infty$.

The construction of the database for vision applications presumes a set of entities and some procedure to derive feature vectors to represent these entities. Typically, the entities to be represented by nodes in the database are examined, and salient features are selected. Each class of feature values is presented as a vector; e.g., a color descriptor could include intensity values obtained from red, green, and blue filtered images. Similarly, shape descriptors can consist of a vector of Fourier coefficients. In this paper, we will use Fourier coefficients as shape descriptors for external contours, and will not consider color or other features.

The domain assumed for the database consists of those problems in which each entity in the database is represented by a vector having d feature values, and corresponding features have their feature values in corresponding locations in the feature vectors. In addition, we assume that the features of the entities are such that taking the difference of corresponding feature values (as a part of applying the L_2 norm) is appropriate. To begin the process of generating a PFNET, it is necessary to compute a scalar weight matrix W . For the purposes of this paper, we will compute the scalar weight values of W using the L_2 norm. For the general case,

$$w_{ij} = \left[\sum_{k=1}^d |w_{ik} - w_{jk}|^m \right]^{1/m}$$

The L_1 norm sums the magnitudes of the differences between corresponding components for the feature vectors being compared, and the L_2 norm is the Euclidean metric. For a discussion of distance measures suitable for data such as this, see Tversky and Krantz (1970). This article also justifies the L_m norms (the Minkowski metrics) as being the only metrics which possess both intra-dimensional subtractivity and interdimensional additivity, a feature that seems as important for vision databases as for cognitive modeling.

PFNETs were the predecessors of empty neighborhood networks (abbreviated ENNs, defined and discussed in Dearholt, et al., in preparation). The distinguishing characteristic of empty neighborhood networks is that each pair of nodes (points) defines a region, or neighborhood, in a general sense. The edge membership rule of these networks requires this neighborhood (for a given pair of nodes) to be empty of other nodes in the set to establish an edge between those nodes. That is, an edge is established between two nodes if and only if the neighborhood defined by those two nodes is empty of other nodes. This can be viewed as each pair of nodes defining a neighborhood which corresponds, in some way, to a "region of influence"; in this region, an edge is established if no other node intrudes to establish its own region of influence. Thus

The reason that PFNET($L2, \infty, 2$) can be efficient in searching for a matching entity in the database is that the edge placement is such that backtracking is rarely needed, provided that there is a matching node in the database. Figure 1 below shows the exceptional case in which backtracking is necessary for the PFNET; the lune between the initial node and the goal node is not empty, so that the two nodes are not linked, and the shortest path spirals (momentarily) away from the goal node. The use of an MSNET for data organization corrects this problem by providing the most economical edges which avoid such spiral paths.

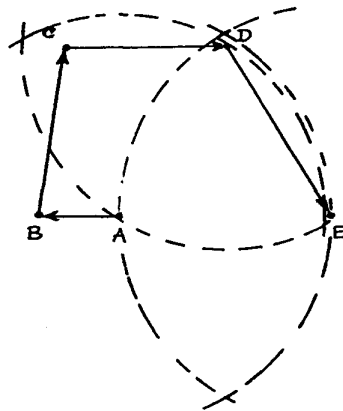


Figure 1. Nonmonotonic Search Path

The generation procedure for MSNETs assumes that $\text{PFNET}(L2, \infty, 2)$ is available. We require that a minimum number of links be added to the PFNET to make it an MSNET. Also, the links added should be the cheapest possible. Recall that an MSNET is a network in which there is a *monotonic* path between every pair of nodes. For any node pair (N_i, N_j) , if $\text{lune}(N_i, N_j)$ is empty, then the edge e_{ij} is present in $\text{PFNET}(L2, \infty, 2)$. In this case, there is a direct path between N_i and N_j and the monotonic search requirement is satisfied. Therefore all links of $\text{PFNET}(L2, \infty, 2)$ are in the MSNET. For any node pair (N_i, N_j) , we order the neighbors of N_j and start with the nearest neighbor and progress to N_i , building monotonic paths from each of the nodes passed (say N_k) to N_j . All paths terminating at node N_j are guaranteed to be monotonic by this process. If edge e_{ik} was established for making the path $P(N_i, \dots, N_j)$ monotonic, and then if later on, e_{ij} was added to make some path $P(N_i, \dots, N_j)$ monotonic, then e_{ik} may be discarded. Now we describe the MSNET generation algorithm more formally:

1. Begin with $\text{PFNET}(L, \infty, 2)$;
2. Create a sorted nearest-neighbor list for each node;
3. Select N_j as the new destination node; if no more nodes are left, then go to step 11;
4. Select N_i , the next nearest neighbor of N_j , as the source node; if no more neighbors are left then go to step 10;
5. If N_i and N_j are directly linked in the MSNET, return to step 4;
6. Pick the next intermediate node N_k (of N_i) from the nearest neighbor list of N_j ; if no more intermediate nodes are left then go to step 9;
7. If $w_{ij} = w_{kj}$ (i.e., duplicate weights), return to step 6;
8. If N_i and N_k are directly linked, continue with step 9; otherwise return to step 6;
9. If N_i was not found to be linked with any of the intermediate nodes, then link N_i to that intermediate node which reduces the distance to N_j and also provides the cheapest connection; return to step 4;
10. Return to step 3;
11. Optimize by discarding redundant edges.

We will now describe the search procedure which we have implemented for searching MSNETs. Assume that a set of entities has been selected and their feature vectors determined, and that an MSNET has been constructed. Also assume that a commensurable external entity X (to be identified) has been selected and that its feature vector has been determined; then the search procedure is as follows:

1. Select an initial node N_i from which to begin the search, and measure the distance $d(X, N_i)$ using the L_2 norm;
2. Using the L_2 norm, measure the distance between X and the nodes linked to N_i , and select the node N_j having the smallest distance $d(X, N_j)$;
3. If $d(X, N_j) < d(X, N_i)$, traverse the edge to N_j and return to step 2;
4. If $d(X, N_j) \geq d(X, N_i)$, apply the match criterion to N_i .
5. If the match is satisfactory, then N_i and X are equivalent and the search halts with a successful match.
6. If the match criterion shows that the match is not satisfactory, then X is not represented in the database, and the procedure halts.

III. Design and Organization of the Database System

The database system is now organized as five modules. These modules have been developed at different stages as the system has evolved. The chart in Figure 2 shows the interaction between the five modules.

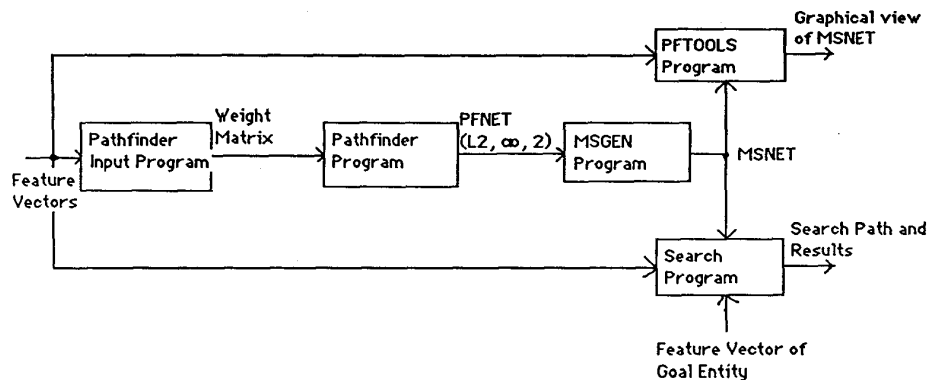


Figure 2. System Organization for the Database

Brief descriptions of the modules follow:

1. Pathfinder Input Program:

The number of nodes and the feature vectors are the inputs to the program. It computes the weight matrix using the L_2 norm. The output is in the format needed by the Pathfinder program.

2. Pathfinder Generation Program:

The Pathfinder generation program, documented in Gerber, et al. (1987), receives comments, number of nodes, minimum weight, maximum weight, and the weight matrix as input. This version produces $PFNET(L_2, \infty, 2)$ as its output.

3. MSGEN Program:

The MSNET generation program takes the output of Pathfinder (number of nodes, weight matrix, and PFNET matrix) as its input and produces the MSNET in matrix form, using the algorithm given in the previous section.

4. Search Program:

The inputs to this program consist of the original feature vectors, the MSNET, and the feature vector of the entity to be searched. The search program currently considers each node in turn, as the *initial node* from which to begin search, as a means of verifying that the database supports monotonic search between each pair of nodes in this small database. The use of heuristics to select an initial node for search has not been developed yet. The present match criterion simply tests for equality, since we have thus far only tested for inclusion or exclusion. A more realistic match criterion, based upon the dimension of the region surrounding each node, is under development. The search paths and whether the search was successful or not are the outputs provided by this program.

5. PFTOOLS Program:

This program (Esposito, 1988) reads in a file of node labels, the feature vectors, and the MSNET to produce a graphical view of the MSNET.

Initial testing of these five modules was done with small sets of data. The major tests done on the integrated system thus far have been with a database of thirty geometric objects (described in Section IV). In addition to the thirty objects in the database, eight archeological objects not present in the database were also used as search goals.

IV. The Objects and Their Organization as an MSNET

The database for this study consists of 30 geometric objects (triangles, rectangles, circles, and other polygons) and simple tools (pliers and wrenches). Additionally, we selected 8 archeological artifacts (primitive tools and projectile points) for testing against the database of 30 objects. Images of all the objects were digitized, and a contouring algorithm was applied to obtain a 4-direction chain code representation of the exterior contour of each object. The chain codes consist of a series of 0's, 1's, and -1's, where a 0 represents a continuation in the same direction, a 1 indicates a 90-degree counterclockwise change of direction, and a -1 indicates a 90-degree clockwise change of direction. The average chain code length for the 30 database objects was 380, the minimum length was 70, and the maximum length was 988. For this study we were interested in shape comparisons only, so that for each contour a normalized arclength of 2π was used to compute the Fourier transform. For a more detailed explanation, see Phillips, Gonzales, Graham, Lopez (1987) and Phillips and Gonzales (1988). Twenty-four Fourier coefficients were computed for each contour, indexed -12 through 12, excluding 0. Each coefficient is a complex

number. The coefficients provide a representation of the two-dimensional shape of an object without loss of information and result in a significant reduction in data storage as compared with storing the images of the chain codes.

When comparing shapes of objects, it is important to have a uniform orientation on which to base comparisons regardless of the orientation of the objects in the input image. For each contour, the Fourier transform was used to find the axis of minimal asymmetry; i.e., the line going through the centroid of the contour about which the contour is the most symmetric (Phillips, Lopez, Gonzales, and Graham, 1987). Fourier coefficients for the contours were recomputed after rotating the original contours to align the axis of minimal asymmetry with the x-axis and to orient the axis of minimal asymmetry so that the distance from the centroid to the contour along the axis of minimal asymmetry was maximized. The normalized Fourier coefficients, 24 complex numbers, are the feature vectors stored for each object. Figure 3 shows the contours of the 30 objects in the database, plotted from the normalized Fourier coefficients. Figure 4 shows the contours of the 8 archeological objects used for testing.

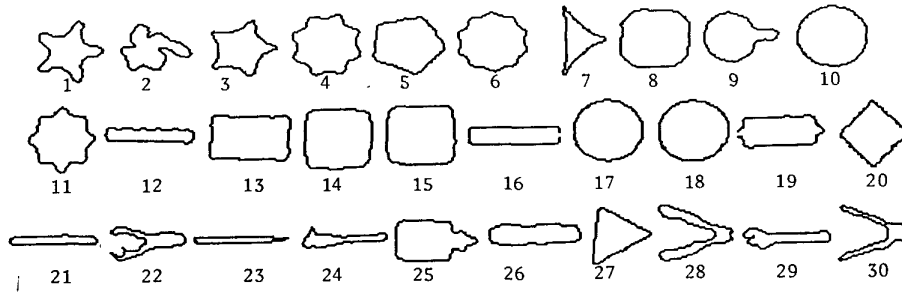


Figure 3. Database Objects



Figure 4. Eight Archeological Test Objects

To generate the components of the weight matrix W , the difference in the shape of each pair of objects is represented by computing the L_2 norm using the normalized Fourier descriptor for each object. Let $\hat{z}$ and $\hat{w}$ be the normalized Fourier transforms of two contours. Then the L_2 norm for them is computed as

$$\|\hat{z} - \hat{w}\| = \left[\sum_{n=1}^N |\hat{z}(n) - \hat{w}(n)|^2 \right]^{1/2}$$

This value becomes a weight in the weight matrix W , and is a good measure of the similarity of the two contours (Phillips, Gonzales, Graham, and Lopez, 1987). The basis for using the L_2 norm to measure the similarity of two contours directly from the Fourier coefficients is the Plancherel theorem for the circle, also called Parseval's identity (Hewitt and Stromberg, 1965). The weight matrix W is used as input to the Pathfinder network generation procedure.

The organization of objects in the database is given by the MSNET in Figure 5. Note the clustering of similar shapes, as indicated by the edges and the positioning of the nodes. The MSNET in Figure 5 is, of course, a two-dimensional representation of a network in 24-dimensional space, so the relative positions of the nodes can not be completely accurate.

V. Results and Conclusions

To assure that the database of 30 objects provided for monotonic search for entities within the database, each node (object) served both as an initial node and as a goal node, for every other node in the database. That is, the search procedure was invoked using each node as an initial node, for every other node as a goal node. In every case, the search was monotonic (no backtracking was required), and the simple match criterion we are using identified the goal node as the one found by the search procedure.

Attempting to find a match for the eight archeological objects in the database (they were not included in the database) proved more interesting. Applying the search procedure, the search path sought the area of similarly-shaped objects, but did not necessarily converge to the node providing the closest match to the test object. The reason for this is that the initial node determines which side of

the (empty) space corresponding to the test object is approached, since edges are followed in the search procedure. Thus when the search becomes nonmonotonic, it just means that the locus of the test object can not be approached closer with the edge structure of the network. We are now investigating the use of the distance information obtained during the search for objects which are not in the database to provide for the efficient addition of nodes to the MSNET, whenever that is desired. When nodes are added, both the properties of monotonic search and the clustering associated with Pathfinder networks and MSNETs must be preserved. Indeed, to be an MSNET, it is possible that some edges may have to be deleted when a node is added, because they may be superseded by other nodes which provide for monotonic search at a lower cost. Having too many edges does not present a problem for monotonic search, as long as the "right" edges are present; but for the descriptive aspects of the system, it may at least be important to label redundant edges.

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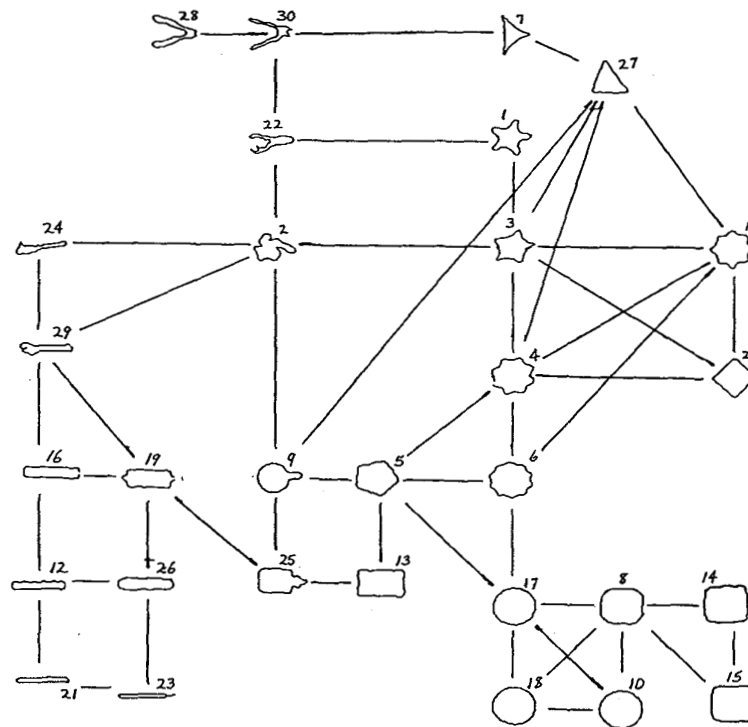


Figure 5. MSNET Organization of the Data

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